

COMPUTER NETWORKS LAB ASSIGNMENT 2

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TCP

Transmission control protocol that runs on top of IP and adds reliability guaranteeing that packets arrive, in order and aren't duplicated.

Packet Delivery Ratio (PDR):

Definition: The percentage of packets successfully delivered to the destination compared to the total number of packets sent by the source.

Formula: $PDR = (\text{Packets Received} / \text{Packets Sent}) \times 100$

Packet Loss Ratio (PLR):

Definition: The percentage of packets that were sent but not received (lost in the network due to drops).

Formula: $PLR = ((\text{Packets Sent} - \text{Packets Received}) / \text{Packets Sent}) \times 100$

Example) for 586 Packets sent, 563 Packets received, 0.5Mb bandwidth and queue size=5:

$PLR = (586 - 563) / 586 = 23 / 586 = 3.92$

And

$PDR = 563 / 586 = 96.08$

FTP:

File transfer protocol designed for transferring files over a TCP/IP network.

Experimenting with change in bandwidth and Queue size:

Bandwidth	Queue Size	Packets Sent	Packets Received	PDR (%)	PLR (%)
0.5 Mb	5	586	563	96.08	3.92
0.5 Mb	10	594	577	97.14	2.86
0.5 Mb	15	602	587	97.51	2.49

1 Mb	5	1116	1082	96.95	3.05
1 Mb	10	1157	1137	98.27	1.73
1 Mb	15	1170	1158	98.97	1.03
2 Mb	5	2023	1986	98.17	1.83
2 Mb	10	2181	2158	98.95	1.05
2 Mb	15	2239	2227	99.46	0.54

Why does packet loss occur in TCP?

Packet loss in TCP mainly occurs due to queue overflow at routers when the sending rate exceeds link capacity. Other causes include congestion on shared links, transmission errors (in wireless), and deliberate early drops by routers. TCP uses these losses as a signal to slow down and avoid further congestion.

Effect of Queue Size and Bandwidth

- Queue Size:

A small queue overflows quickly

→ more packets are dropped

→ lower PDR and higher PLR.

A larger queue can hold more packets

→ fewer drops

→ higher PDR and lower PLR.

- Bandwidth:

Low bandwidth means the link transmits fewer packets per second

→ congestion builds up faster

→ more drops.

Higher bandwidth can transmit packets faster

→ reduces congestion

→ improves PDR and lowers PLR.